We have computed the Lie bracket for $\mathfrak{sl}(2,\mathbb{C})$ using the vector field commutator.

Define $X_i = \left(\frac{\partial}{\partial x^i}\right)_{id}$

$$[X_1, X_2] = 2X_2 \;;\; [X_2, X_3] = X_1 \;;\; [X_3, X_1] = 2X_3$$

General

$$[X_i, X_j] = C^k{}_{ij} X_k \Rightarrow C^2{}_{12} = 2 = C^3{}_{31} \;;\; C^1{}_{23} = 1.$$

- Killing form of $\mathfrak{sl}(2,\mathbb{C})$.

$$K(A,B) = \mathrm{tr}(\mathrm{ad}(A) \circ \mathrm{ad}(B))$$

$$K(X_i, X_j) = K_{ij} = C^m{}_{in} C^n{}_{jm} \quad \text{symmetric matrix.}$$

$$= C^n{}_{jm} C^m{}_{in}$$

$$= C^m{}_{jn} C^n{}_{im} = K_{ji}$$

$$K_{11} = C^m{}_{1n} C^n{}_{1m} = C^2{}_{12} C^2{}_{12} + C^3{}_{13} C^3{}_{13} = 8.$$

$$K_{22} = C^m{}_{2n} C^n{}_{2m} = C^m{}_{21} C^1{}_{2m} + C^m{}_{23} C^3{}_{2m} = 0$$

$$K_{33} = C^m{}_{3n} C^n{}_{3m} = 0.$$

$$K_{12} = C^m{}_{1n} C^n{}_{2m} = C^m{}_{11} + C^2{}_{12} C^2{}_{22} + C^3{}_{13} C^3{}_{23} = 0.$$

$$K_{13} = C^m{}_{1n} C^n{}_{3m} = C^3{}_{13} C^3{}_{33} + C^2{}_{12} C^2{}_{32} = 0$$

$$K_{23} = C^m{}_{2n} C^n{}_{3m} = C^2{}_{21} C^1{}_{32} + C^1{}_{23} C^3{}_{31} = 4.$$

$$(K_{ij}) = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}$$

Recall Cartan Criteria: $(2^{nd}) \to (L, [\cdot,\cdot])$ Lie algebra is semi-simple iff K is non-degenerate $\Leftrightarrow \det(K_{ij}) \neq 0$.

$\det(K_{ij}) = -128 \neq 0. \Rightarrow \mathfrak{sl}(2,\mathbb{C})$ is semi-simple.

$\Rightarrow L_1 \oplus L_2 \oplus \cdots \oplus L_n$ (simple)

If G is a compact Lie group, then K negative definite. ⇝ Need Haar Measure.

as K is not negative definite $\Rightarrow SL(2,\mathbb{C})$ is not compact.

- Claim: $\mathfrak{sl}(2,\mathbb{C})$ is simple

Recall: L is simple, if it contains no non-trivial ideals.

$I \subseteq L$ (subspace) if $[I, L] \subseteq I$. $\{0\}$ & L are ideals trivial.

We try to construct an non-trivial ideal explicitly

$$I = \left\{ \alpha X_1 + \beta X_2 + \gamma X_3 \;\middle|\; \begin{array}{l}\alpha,\beta,\gamma \text{ s.t.} \\ \text{we don't have} \\ \text{a trivial ideal}\end{array} \right\} \;;\; (\alpha,\beta,\gamma \in \mathbb{C}) \times$$

$(\alpha = \beta = \gamma = 0) \times$

Since I is subspace itself, if $[\![I, X_i]\!] \subseteq I$ for all i, we are done

$\Rightarrow$ only need to compute $[\![\alpha X_1 + \beta X_2 + \gamma X_3, X_i]\!]$

$\Rightarrow$ $i=1 \Rightarrow [\![\alpha X_1 + \beta X_2 + \gamma X_3, X_1]\!] = -2\beta X_2 + 2\gamma X_3$

$i=2 \Rightarrow [\![\alpha X_1 + \beta X_2 + \gamma X_3, X_2]\!] = 2\alpha X_2 - \gamma X_1$

$i=3 \Rightarrow [\![\alpha X_1 + \beta X_2 + \gamma X_3, X_3]\!] = -2\alpha X_3 + \beta X_1$

If $\alpha \neq 0$, $\beta \neq 0$; $\gamma \neq 0$ we got three linearly independent vectors.

Let $Y \in \mathfrak{sl}(2,\mathbb{C})$, $Y = y^i X_i$

$[\![I, L]\!] \subseteq I \Rightarrow \boxed{I = L}$ ← we don't want this.

Let's start relaxing the conditions $\Rightarrow$ atleast one of α, β, γ must be zero.

⊙ $\underline{\alpha = 0}$: $[\![\ ,\ ]\!] \subset \text{span}(\{X_2, X_3\})$

$\Rightarrow \gamma = 0$ & $\beta = 0$ $\Rightarrow$ $I = \{0\}$.

⊙ $\underline{\beta = 0}$: $[\![\ ,\ ]\!] \subset \text{span}(\{X_1, X_3\})$

$\Rightarrow \alpha = 0 \Rightarrow [\![\ ,\ ]\!] \subset \text{span}(\{X_3\}) \Rightarrow \gamma = 0$ $\Rightarrow I = \{0\}$.

⊙ $\underline{\gamma = 0}$: $[\![\ ,\ ]\!] \subset \text{span}(\{X_1, X_2\})$

$\Rightarrow \alpha = 0 \Rightarrow [\![\ ,\ ]\!] \subset \text{span}(\{X_2\}) \Rightarrow \beta = 0$ $\Rightarrow I = \{0\}$.

$\rightarrow$ $\nexists$ non-trivial ideal of $\mathfrak{sl}(2,\mathbb{C})$.

$\Rightarrow$ $\mathfrak{sl}(2,\mathbb{C})$ is simple.

- $\underline{\text{Roots}}$ and $\underline{\text{Fundamental roots}}$ of $\mathfrak{sl}(2,\mathbb{C})$.

Start by finding the Cartan subalgebra, H

$\underline{\text{Claim}}$: $H = \text{span}_{\mathbb{C}}\{X_1\}$.

basis extension is obv. $\{X_1, X_2, X_3\}$.

$\text{ad}(X_1)X_2 = [\![X_1, X_2]\!] = 2X_2$

$\text{ad}(X_1)X_3 = [\![X_1, X_3]\!] = -2X_3$

$\Rightarrow$ they are indeed eigenvectors.

$\Rightarrow$ H is Cartan subalgebra.

define $\lambda_2 : H \xrightarrow{\sim} \mathbb{C}$

$\forall h \in H : \exists \xi \in \mathbb{C} : h = \xi X_1$

$\Rightarrow \lambda_2(h) = \xi \lambda_2(X_1)$

$\lambda_2(X_1) = 2$, $\lambda_3(X_1) = -2$ } roots.

and observe $\boxed{\lambda_2 = -\lambda_3}$

$\hookrightarrow$ $\lambda_2, \lambda_3 \in H^*$. $\Phi = \{\lambda_2, \lambda_3\}$.

- $\underline{\text{fundamental roots}}$: $\Pi \subset \Phi$

s.t. (a) Π is linearly indepedent

(b) $\forall \lambda \in \Phi$ $\exists n_1, \dots, n_{|\Pi|} \in \mathbb{N}_0$ $\exists \varepsilon \in \{+1, -1\}$:

$\lambda = \varepsilon \sum_{i=1}^{|\Pi|} n_i \pi_i$.

Here: $\Pi = \{\lambda_2\}$.

– Weyl transformations

$$S_\lambda(\mu) := \mu - 2\frac{K^*(\lambda,\mu)}{K^*(\lambda,\lambda)}\lambda \quad ; \quad K^*: H_R^* \times H_R^* \to \mathbb{R} \quad \text{with} \quad K^*(\mu,\nu) = K\left(i^{-1}(\mu), i^{-1}(\nu)\right)$$

$\lambda \in \Phi$ and $\mu \in H^*$; $H_R^* = \text{span}_R(\Pi)$; $i(h) \in H^*$; $i(h)(g) = K(h,g) \in \mathbb{C}$.

↳ $S_{\lambda_2}(\lambda_2) = \lambda_2 - 2\frac{K^*(\lambda_2,\lambda_2)}{K^*(\lambda_2,\lambda_2)}\lambda_2 = -\lambda_2$

$\Rightarrow \boxed{W(\Pi) = \Phi}$.

$W = \{S_\lambda \mid \lambda \in \Phi\}$; Weyl group.

★ Cartan Matrix: $C = (2)$; Dynkin Diagram: $\mathfrak{sl}(2,\mathbb{C}) \longrightarrow$ O $\longrightarrow A_1$

This completes our analysis for Lie group $SL(2,\mathbb{C})$ and Lie algebra $\mathfrak{sl}(2,\mathbb{C})$, starting from its set def.

→ Now, we will work on another example.

Reconstruction of a Lie Algebra from its Dynkin diagram

Consider A_2 : ① —— ②

$\Pi = \{\pi_1, \pi_2\}$; bond no. $n_{12} = 1$.

$n_{ij} = C_{ij}C_{ji} \Rightarrow 1 = C_{12}C_{21} = 4\cos^2\varphi_{ij}$

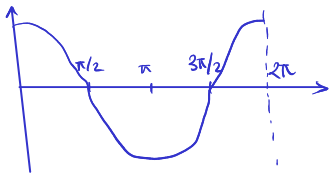
$\Rightarrow |\cos\varphi_{12}| = 1/2$.

$\Rightarrow \boxed{\varphi_{12} = 2\pi/3}$

$C_{ij} = 2\frac{K^*(\pi_i,\pi_j)}{K^*(\pi_i,\pi_i)}$ (< 0 for $i \neq j$)

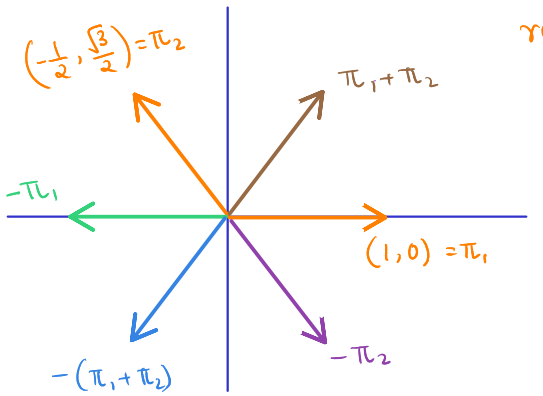
$\Rightarrow K^*(\pi_1,\pi_2) < 0$

$\Rightarrow \varphi_{12} > \pi/2$.



$$\Pi \subset \Phi \subset \text{span}_{\mathbb{Z}\in\mathbb{N}_0}(\Pi) \subset \underbrace{H_R^* = \text{span}_R(\Pi)}_{\text{all the bond no. and angles are for this level.}} \subset H^* = \text{span}_{\mathbb{C}}(\Pi).$$

⇒ Picture in H_R^*.



repeated action of Weyl group:

- $S_{\pi_1}(\pi_1) = -\pi_1$
- $S_{\pi_1}(\pi_2) = \pi_2 - 2\frac{K^*(\pi_1,\pi_2)}{K^*(\pi_1,\pi_1)}\pi_1 = \pi_2 - 2\frac{(-1/2)}{1}\pi_1 = \pi_2 + \pi_1$
- $S_{\pi_2}(\pi_2) = -\pi_2$
- $S_{\pi_2}(-\pi_1) = -\pi_1 - 2\frac{K^*(\pi_2,-\pi_1)}{K^*(\pi_2,\pi_2)}\pi_2 = -\pi_1 - 2\frac{(1/2)}{1}\pi_2 = -(\pi_1+\pi_2)$

any other action we now only produce a element from this.

$\Rightarrow |\Phi| = 6$. and $|\Pi| = 2$.

⊛ $\dim_{\mathbb{C}} H = \dim_{\mathbb{C}} H^* = \dim_{\mathbb{R}} H^*_{\mathbb{R}} = |\Pi| = 2.$

$$\Rightarrow \dim L = \dim_{\mathbb{C}} H + |\Phi|$$

$$\Rightarrow \dim L = 6 + 2 = 8.$$

To fully reconstruct the algebra, we need to compute $[\![\cdot,\cdot]\!]$ or $C^k{}_{ij}$.

Cartan-Weyl Basis:

$$\{h_1, h_2, e_1, \dots, e_6\}$$

$H = \mathrm{span}_{\mathbb{C}}(\{h_1, h_2\})$.

$\forall h \in H$: $\mathrm{ad}(h)e_\alpha = \lambda_\alpha(h)e_\alpha$

$$\Downarrow$$

$$[\![h_A, e_\alpha]\!] = \mathrm{ad}(h_A)e_\alpha = \lambda_\alpha(h_A)e_\alpha \quad \begin{matrix} A = 1, 2 \\ \alpha = 1, \dots, 6 \end{matrix}$$

Since A_2 is simple, $\Rightarrow H$ is abelian

$$\Rightarrow [\![h_1, h_2]\!] = 0 \Rightarrow \boxed{C^k{}_{12} = C^k{}_{21} = 0}, \quad k = A, \alpha.$$

⊛ Using $\mathrm{ad}(h_A)e_\alpha = [\![h_A, e_\alpha]\!] = \lambda_\alpha(h_A)e_\alpha$

$$\Rightarrow \boxed{\begin{aligned} C^\alpha{}_{A\alpha} &= \lambda_\alpha(h_A) \\ C^B{}_{A\alpha} &= 0. \end{aligned}} \quad \forall A = 1, 2, \ \alpha = 1, \dots, 6 \quad [\text{No-summation}].$$

Here, λ_α depends our choice, how we label roots generated by action of Weyl group.

✳ example, label root π_1, for basis e_1.

$$[\![h_1, e_1]\!] = \lambda_1(h_1)e_1 \quad ; \quad \text{Recall, } \pi_1 = (1, 0) \Rightarrow \lambda_1(h_1) = 1, \ \lambda_1(h_2) = 0.$$

and so on.

e_1	e_2	e_3	e_4	e_5	e_6
λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
π_1	π_2	$\pi_1 + \pi_2$	$-\pi_1$	$-\pi_2$	$-(\pi_1 + \pi_2)$

These covers all the scenerios for $[\![h_A, h_B]\!]$, $[\![h_A, e_\alpha]\!]$, we need $[\![e_\alpha, e_\beta]\!]$.

For this we'll we Jacobi identity,

$$[\![h_A, [\![e_\alpha, e_\beta]\!]]\!] = -[\![e_\alpha, [\![e_\beta, h_A]\!]]\!] - [\![e_\beta, [\![h_A, e_\alpha]\!]]\!]$$

$$= -[\![e_\alpha, -\lambda_\beta(h_A)e_\beta]\!] - [\![e_\beta, \lambda_\alpha(h_A)e_\alpha]\!]$$

$$\mathrm{ad}(h_A)[\![e_\alpha, e_\beta]\!] = \big(\lambda_\alpha(h_A) + \lambda_\beta(h_A)\big)[\![e_\alpha, e_\beta]\!]$$

$$= \underbrace{(\lambda_\alpha + \lambda_\beta)}(h_A)[\![e_\alpha, e_\beta]\!] \Big\} \longrightarrow \text{eigenvalue eqn.}$$

↳ something like another root.

And roots generates all the possible eigenvalues

⊛ for e_1, e_2

$$\mathrm{ad}(h_A)[\![e_1, e_2]\!] = \underbrace{(\pi_1 + \pi_2)}(h_A)[\![e_1, e_2]\!]$$

$$\Vert$$

$$\lambda_3 \rightsquigarrow e_3$$

$$\Rightarrow \boxed{[\![e_1, e_2]\!] = \xi e_3} \quad \text{for some } \xi \in \mathbb{C}.$$

Thus, from the example, we can conclude that, if $\lambda_\alpha + \lambda_\beta \in \Phi$

then $[\![e_\alpha, e_\beta]\!] = \xi e_\gamma$ for some $\gamma = 1, \dots, 6$.

Now, it may happen that $\lambda_\alpha + \lambda_\beta \notin \Phi$, then we have two possible case, as to satisfy the eqn, both sides should vanish.

- $\lambda_\alpha + \lambda_\beta \notin \Phi$ and $\lambda_\alpha + \lambda_\beta \neq 0$.

  $\Rightarrow [\![e_\alpha, e_\beta]\!] = 0$.

- $\lambda_\alpha + \lambda_\beta \notin \Phi$ and $\lambda_\alpha + \lambda_\beta = 0$.

  $\Rightarrow [\![e_\alpha, e_\beta]\!] \in H$. } $\rightarrow$ H is maximal abelian

  $\Rightarrow x \in L : \forall h \in H : [\![h, x]\!] = 0 \Rightarrow x \in H$.

$$\rightarrow [\![e_\alpha, e_\beta]\!] = \begin{cases} N_{\alpha,\beta} e_\gamma, & \lambda_\alpha + \lambda_\beta \in \Phi \\ 0, & (\lambda_\alpha + \lambda_\beta \neq 0) \notin \Phi \\ h_\alpha \in H, & \lambda_\alpha + \lambda_\beta = 0 \end{cases}$$